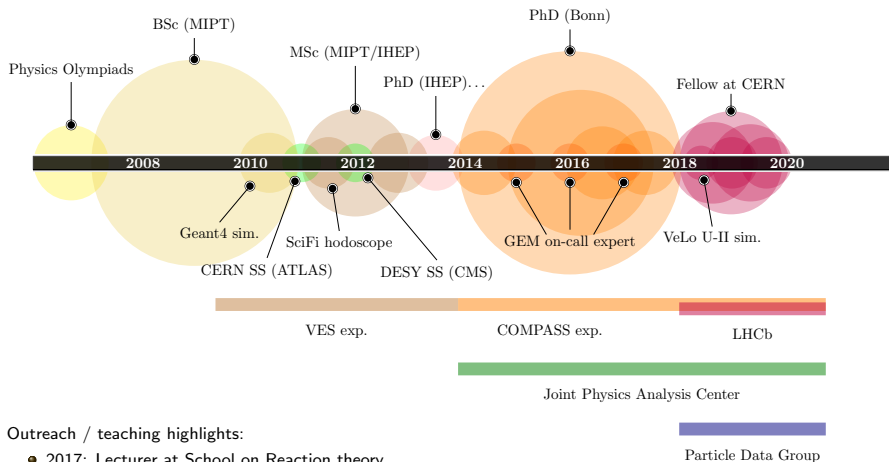


Mikhail Mikhasenko: career path and projects

<https://github.com/mmikhasenko>



Outreach / teaching highlights:

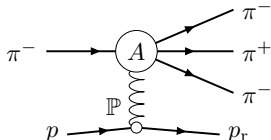
- 2017: Lecturer at School on Reaction theory
- CERN learning initiative (SS 2020): workshop on amplitude analysis
- COMPASS, LHCb guide

Search for exotic hadrons and hadronic parameters

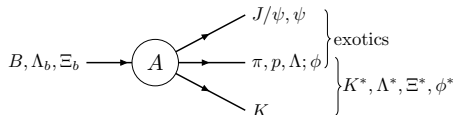
The need for the Amplitude Analysis

Understanding QCD at low energy: resonance physics, matter formation, input to CPV

In scattering (COMPASS/VES):



In decays (LHCb):

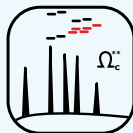
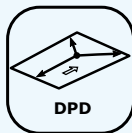


- Describe structures in the spectrum
 - identify conventional states
 - kinematic effects
 - distinguish from the cross-channel interaction

⇒ Amplitude Analysis

My contribution is filling the gap between theory and experiment.

Highlights of my LHCb projects

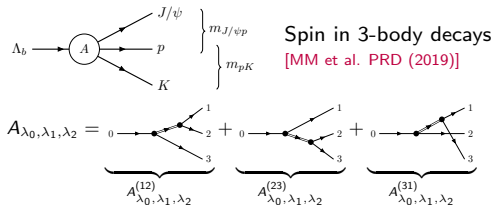


hardware: SiPM shaper, GEM, VeLo

future plans: $X(3872)$, T_{cc} , H_c , QTI, ...

Highlights of my recent work

Dalitz Plot Decomposition (DPD)



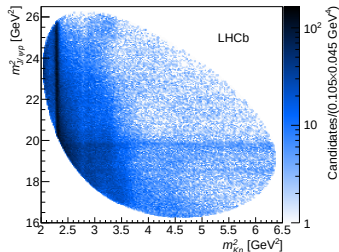
Used in the past

$$A_{\lambda_0, \lambda_1, \lambda_2}^{(23)} = \underbrace{\boxed{\phi_i, \theta_i}}_{0 \rightarrow X, 1} \times \underbrace{\boxed{\phi'_i, \theta'_i}}_{X \rightarrow 2, 3} \times \underbrace{\boxed{\phi''_i, \theta''_i}}_{\text{spin align.}}$$

- "impossible" to get right
- spin 1/2: $A(\pi) \neq A(-\pi)$
- range of ϕ matters $[-\pi, \pi]$ vs $[0, 2\pi]$

Consequences of the incorrect spin matching:

- Unpolarized decay is not inhomogenous
- Observed intensity is 2π discontinuous in ϕ
- Broad components are impacted the most
- Determination of J^P might be incorrect



Proposed in DPD

$$A_{\lambda_0, \lambda_1, \lambda_2} = \sum_{\nu} D_{\lambda_0, \nu}^{1/2*}(\alpha, \beta, \gamma) \underbrace{O_{\lambda_1, \lambda_2}^{\nu}(m_{12}^2, m_{23}^2)}_{O^{(12)} + O^{(23)} + O^{(31)}}$$

$$O_{\nu, \lambda_1, \lambda_2}^{(23)}(m_{12}^2, m_{23}^2) = \underbrace{\boxed{\phi_i, \theta_i}}_{0 \rightarrow X, 1} \times \underbrace{\boxed{\phi'_i, \theta'_i}}_{X \rightarrow 2, 3} \times \underbrace{\boxed{\phi''_i, \theta''_i}}_{\text{spin align.}}$$

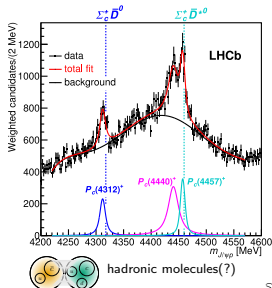
- correct by construction

Impacted analyses:

P_{cs}	$\Xi_b^- \rightarrow J/\psi \Lambda^0 K^-$ just published
Z, X	$B^- \rightarrow J/\psi \phi K^-$ at RC
CPV	$\Xi_b^- \rightarrow K^- p K^-$ at RC
P_c	$B_s^0 \rightarrow J/\psi p \bar{p}$ at RC
Z	$B^0 \rightarrow J/\psi (\psi') \pi^+ K^-$
$H_{\lambda, \lambda}$	$\Lambda_c^+ \rightarrow p K^- \pi^+$

Better understanding of the resonance dynamics

Lineshape function is a key ingredient to learn underlying dynamics, and get a hint on microscopic picture.



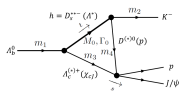
$$\Lambda_b, \Lambda \rightarrow P_c \rightarrow J/\psi, \mu \rightarrow p, \lambda \rightarrow K = \underbrace{\Lambda_b^0 \rightarrow P_c, K^-}_{\text{resonance dynamics}} \times \underbrace{L(m^2)}_{\text{resonance dynamics}} \times \underbrace{P_c \rightarrow p, J/\psi}_{\text{spin align.}}$$

$$L(m^2) \stackrel{?}{=} \frac{1}{M^2 - m^2 - iM\Gamma} \quad (\text{BW approximation(!)})$$

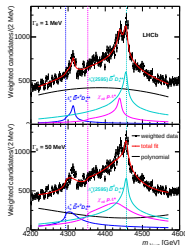
I lead the effort on advanced methods:

- ⇒ Methods based on scattering theory (K, N/D)
- ⇒ Multichannel parametrization
- ⇒ Extraction of T-matrix pole parameters (+PDG)

Investigation of the TS



no acceptable description of P_c in the rescattering picture has been found



Impacted LHCb analyses:

$$P_c \quad \Lambda_b^0 \rightarrow J/\psi p K^- \text{ (publ.)}$$

$$Z, X \quad B^- \rightarrow J/\psi \phi K^- \text{ (in RC)}$$

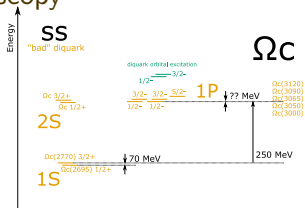
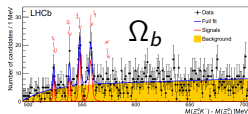
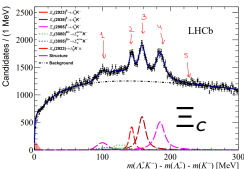
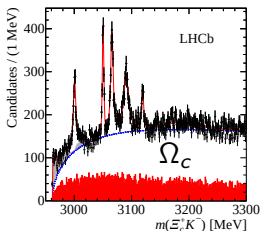
$$X(3872) \quad B^- \rightarrow J/\psi \pi^+ \pi^- K^-$$

$$\Lambda_b^{*0} \quad \Lambda_b^{*0} \rightarrow \Lambda_b^0 \pi^+ \pi^- \text{ (publ.)}$$

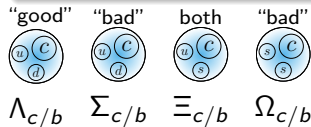
$$X/T_{cc} \quad D\bar{D} / D\bar{D}^* / DD / DD^*$$

Heavy baryons spectroscopy

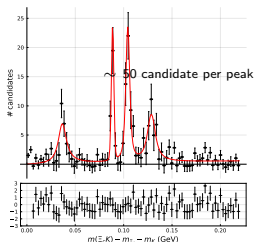
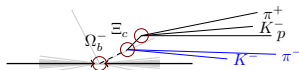
Five lines mystery



5 lines pattern is a feature of the "bad" diquark

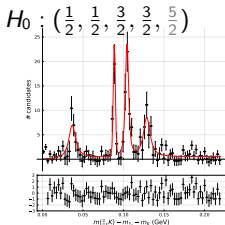
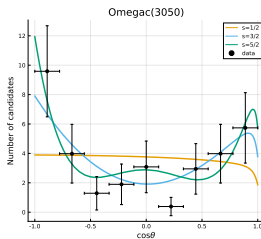
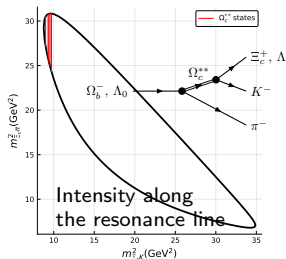


First observation of the bad-diquark 1P multiplet in exclusive reaction



- Selection
- Kinematic fitting
- Efficiency map
- Resolution MC
- FeedException
- Significances
- Test Statistics

Ω_c^{**} analysis



$J \neq 1/2$

- H_0 is excl. with $> 3\sigma$
- Instead, $(\frac{1}{2}, \frac{3}{2}, \frac{3}{2}, \frac{5}{2})$ is supported
- missed $1/2$ state (?): either broad, or below the threshold, or ...

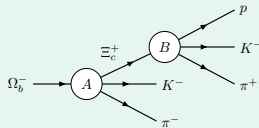
Analysis passed the WG review
On track for Moriond QCD 2021

“Polarization Sensitivity”:

(I just invented the quantity)

is a property of the decay expressing how the final state reflects polarization

- Property of hadron decay
- Can be measured once and transferred to any other decay
- Needs generalization for $3b$ decay



$$M = A(m^2, m^2)D(\alpha, \beta, \gamma)B(m^2, m^2)$$

Analysis of $\Xi_c \rightarrow pK\pi$
CERN SS 2020 project

Addressing the selection criteria

Collaborative work

I consider it essential in the modern experimental physics:

- the key ingredient for addressing large complex problems
- an important part of my strategic planning

My teams:

- **Edinburgh/Bari/CERN @ LHCb**: a small team dedicated to Ω_c^{**} analysis
- **B2PsiX AmAn**: 10 members, 3 LHCb analysis with extensive theory input
- **JPAC**: 6 senior postdoc, junior professors (3 are affiliated with LHCb) – diverse projects on reaction theory
- **Moscow/Novosibirsk/Marseille/Syracuse @ LHCb**: $D^0 D^0 / D^0 \bar{D}^0$ analyses
- **Bonn group @ COMPASS**: co-advising 3 PhD students working on PWA

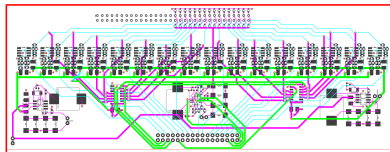
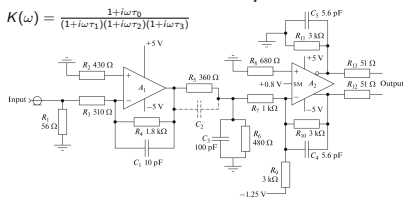
It is critical to me to complement coordination with my own practical contribution!

Work with Manchester group

I am eager to join new collaborative efforts that go beyond my previous experiences

Hardware and other experiences

[Instrum. Exp. Tech. **63** (2020) no.2, 165-182]



I have designed, traced, soldered, and tested the 16-channels shaper board (SiPM, PM).

Commissioning and operation

GEM detectors at COMPASS

- Calibration and maintainance
- On-call duty during 2016-2018 runs

CMS Pixel detector (DESY SS)

- Test beam measurements
- Modeling of the charge distribution

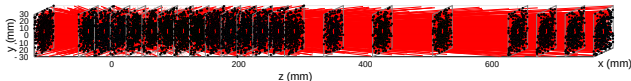
Simulation projects

U-II VeLo: 4d tracker simulations

[VeLo framework TDR (under WG review)]

Geant4 simulations of the Active Target

[my BSc thesis]



Communication skills

Teaching

Regular classes:

- Particle physics, Hadron Physics, Group theory

Specilized courses:

- Amplitudine analysis, Reaction theory
- Physics for medical students

Physics schools:

- 2020: CERN Summer School
- 2018: Hadron Physics Summer School
- 2017: Summer School on the Reaction Theory

Talks with large audience (last 3y)

(fut.) APS GHF
(invited)

(fut.) QWG21

INT20 workshop
(invited)

LHCb implication
(invited)

MIAPP2019

ATHOS2019

BLOIS2019

LHCb-BESIII
(invited)

CHARM2018
(invited)

IWHSS2018
(invited)

Plans and ideas

Future plans

Spin and parity of exotics

P_c, P_{sc}, Z_c, Z_{sc}

Analysis of c/b baryons

$\Lambda_{c/b}^{**}, \Sigma_{c/b}^{**}, \Xi_{c/b}^{**}, \Omega_{c/b}^{**}$

Charmonium exotics

$X(3872), T_{cc}$

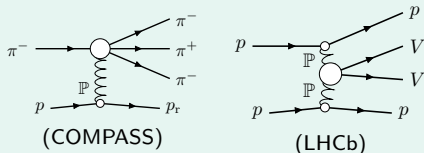
VeLo Upgrade II concept and detailed studies

Exploratory:

CERN Quantum Technology Initiative

- I have initiated a journal club on Th.&Comp. at CERN
- Explore applications of QC at LHCb

From COMPASS diffraction to CEP



[MM et al., arXiv:2007.05501]

“Polarization Sensitivity” for AmAn

- Establish the concept
- Measure $\Lambda_c^+, \Xi_c^+ \rightarrow ph^+ h^-$
- Transfer the knowledge to complex decay chains, e.g. $\Lambda_b^0 \rightarrow \Lambda_c^+ \bar{D}^0 K^-$

[CERN-STUDENTS-Note-2020-031 (advisor MM)]